

Goal: To identify the intentional organization structures of the Enron social network and to be able to decipher the complex social hierarchy from network graph analysis.

On neighbourhood degree sequences of complex networks - Kevin M. Smith

Neighborhood Degree Sequences - Give us an idea about the local structure of a node
We assume that the number of connections is directly proportional to the importance of a node in the network in the context of an organizational hierarchy. (This assumption is not true for all social networks. Hence, NDS is a fair representation of the local structure around a node.

Params Used:

1. Node Heterogeneity

Measure of variance/spread in the degrees of a node's neighborhood in the network on average.

If the spread is less, the neighboring nodes all have similar degrees, or if the spread is high, the neighboring nodes have varying degrees, irrespective of the degree of the node around which the spread is measured.

2. Neighborhood Similarity (S)

Proportion of nodes in the network whose NDS matches that of another node in the graph. S is always between 0 and 1. This metric gives us an idea of the proportion of nodes in the network with a non-unique local structure around them, i.e., this metric can answer the question, are there any repetitions in the local structures around nodes.

3. Hierarchical Complexity

Measures the variation in neighbourhood structures among nodes of the same degree.

4. Network Organization

Measures how structured and organized the network is beyond randomness.

We compare the network graph with random graphs with the same degree distribution because we wish to answer the question

* Are repeated neighborhood patterns intentional structure or just mathematical consequences of degrees?

There is a chance that there is no intention behind the existence of a unique-looking structure owing to randomness. We wish to isolate and avoid drawing any conclusions that lead us to mistake randomness for organisation.

Observed Metric = degree-driven effects(captured by CM) + true organization(what we want)

Configurational Models and Wilcoxon signed-rank test.

With truncation of the feature vector or NDS to 20 degrees and 10 config models.

--- REAL NETWORK ---

--- CONFIGURATION MODELS ---

Run 1/10
Run 2/10
Run 3/10
Run 4/10
Run 5/10
Run 6/10
Run 7/10
Run 8/10
Run 9/10
Run 10/10

--- COMPARISON ---

Vn:

Real: 24931.97767
CM mean: 15024.46580
p-value: 1.95312e-03

S:

Real: 0.56034
CM mean: 0.34372
p-value: 1.95312e-03

Omega:

Real: 0.02229
CM mean: 0.00671
p-value: 1.95312e-03

R:

Real: 145204.38721
CM mean: 31112.81115
p-value: 1.95312e-03

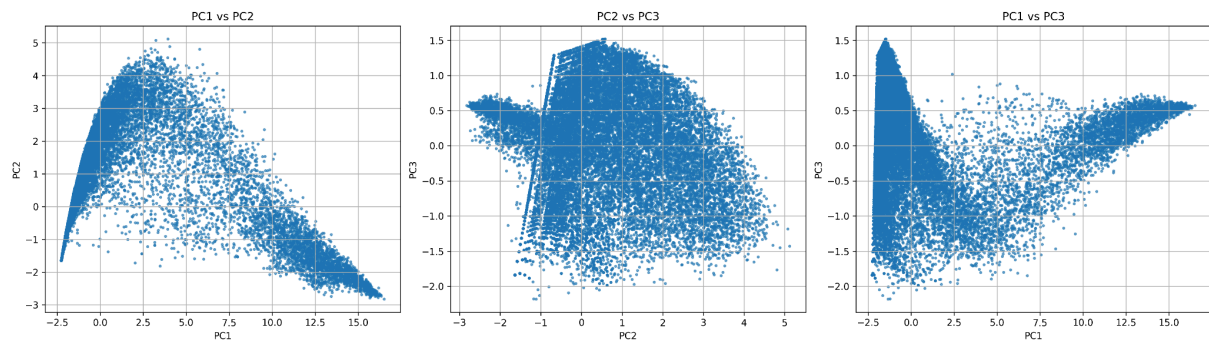
R_omega:

Real: 145201.16610
CM mean: 31112.75320
p-value: 1.95312e-03

At 95% confidence interval, all the metrics reject the null hypothesis that real = cm mean, hence it can be said with 95% confidence that none of the metric the CM mean. Hence, there is more information in the graph that doesn't arise from randomness alone.

Using Principal Component Analysis.

The feature vector for a node is the 20 neighbor nodes with the highest degrees. Using PCA on the data gives us 20 transformed dimensions with approximately 80% variance explained by the first component, PC1 (pca_variance_explained.csv)



#Influence Scoring for a node:

$$\text{influence} = \alpha * \text{degree} + (1-\alpha) * \text{PCA1}$$

degree - factors into account the direct influence of a node due to the number of neighbors it has.

PCA1 - a measure of how highly connected the neighboring nodes of a node are, i.e., the quality of a node's neighbors in the context of influence, and therefore is a factor that needs to be taken into account.

The top 10 nodes with the most influence are:

Node	PC1	Degree	influence_score
273,	15.630801900593204,	1367,	4.648625797026574
136,	16.343684362965224,	1026,	4.637095283835498
1028,	15.783899855229082,	1244,	4.632703819117593
370,	16.107140170925376,	1099,	4.630019537728157
140,	15.756635517885885,	1245,	4.628958702192976
458,	15.701687095995492,	1261,	4.626000142891262
195,	15.856335268161317,	1143,	4.608681748042489
1139,	15.62554621684311,	1068,	4.5462487738122235
566,	15.977418424706109,	924,	4.539252111152948

The influence score model used takes into account influence due to direct connections and indirect connections due to connections of the neighbors.

Another interesting method to classify identical nodes is discussed in the attached research paper (ART_MODEL), wherein a generative probabilistic model called Author-Receipient-Topic model is used to classify nodes as belonging to the same class or based on the people they converse with (receipients) and how they converse with these receipients (i.e., the context of the emails they send).